Phase 3: Relational Model and Normalization

Team number: 32

Assigned Phase 2 submission converted from Team 41 (ACTG), domain ICC Rankings

The EER used is the final Phase 2 diagram of Team 41, together with the corrections the team listed in its report: retired_formats as a composite multi-valued attribute (format_code, retired_on), needs_recompute on TEAM_RATING, and the status attribute on MATCH. The Phase 1 requirements of that domain were used only to interpret the diagram.

Stage A: ER-to-Relational Mapping

The EER diagram becomes 17 relations

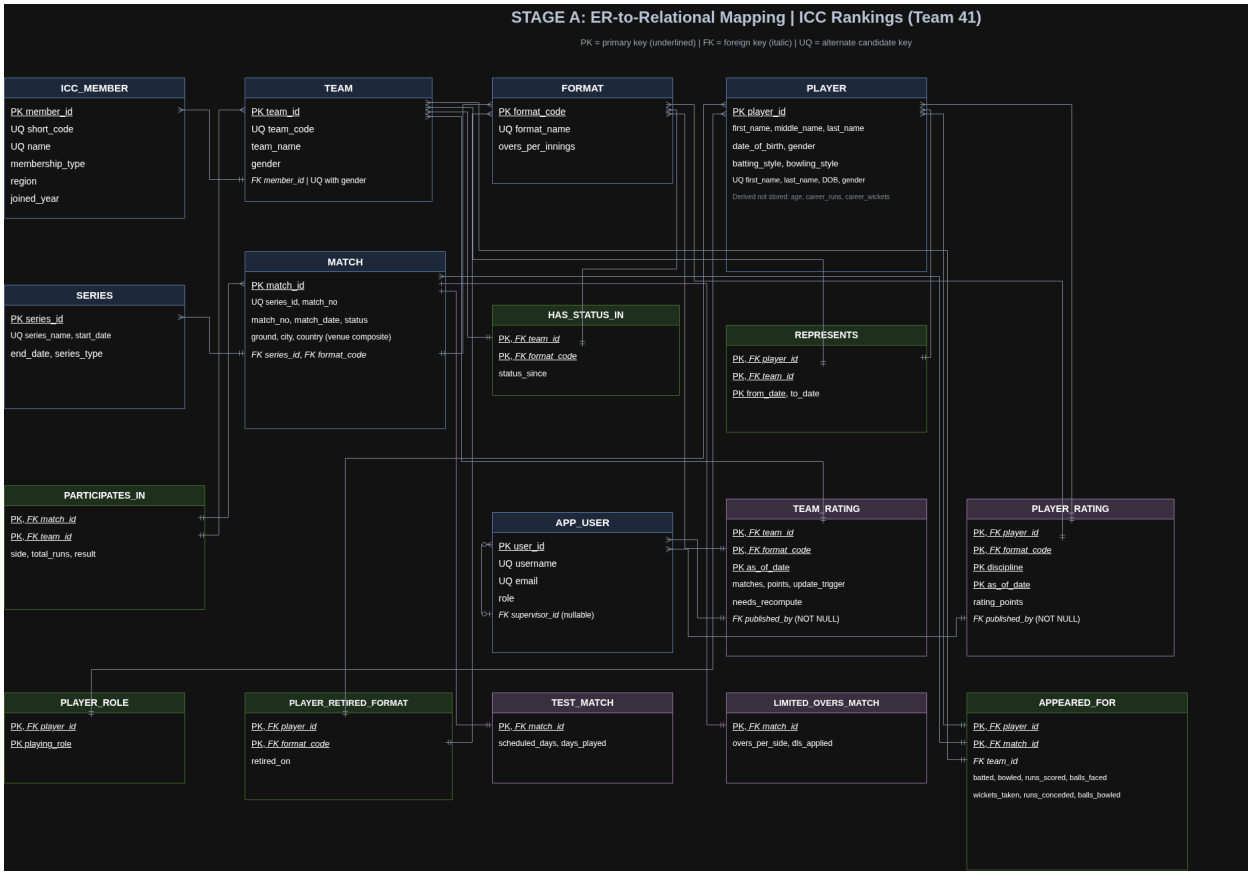
- **Strong entities (step 1):** ICC_MEMBER, TEAM, FORMAT, PLAYER, SERIES, MATCH, APP_USER, each with its simple attributes. The composite full_name of PLAYER became first_name, middle_name, last_name. The composite venue of MATCH became ground, city, country.

  Derived attributes not stored: PLAYER.age, career_runs, career_wickets and last_match_date (computed from date_of_birth, APPEARED_FOR, and the latest match_date of the player's appearances in MATCH), TEAM_RATING.rating (= ROUND(points ÷ matches)) and rank_position, and PLAYER_RATING.rank_position (computed by sorting). The all-rounder index and ranking eligibility are also treated as derived domain information, so they are not stored as separate columns.
- **Weak entities (step 2):** TEAM_RATING has PK (team_id, format_code, as_of_date); team_id is the owner key from TEAM and (format_code, as_of_date) is the partial key. PLAYER_RATING has PK (player_id, format_code, discipline, as_of_date) with owner PLAYER. Both are identifying relationships (TEAM_RATED, PLAYER_RATED), so no extra relation is needed. format_code is also an FK to FORMAT.
- **1:N relationships (step 3):** BELONGS_TO gives TEAM.member_id (NOT NULL FK to ICC_MEMBER because TEAM participation is total). The FK alone does not enforce the requirement that every ICC_MEMBER has at least one TEAM; that needs a separate integrity constraint. CONTAINS gives MATCH.series_id and PLAYED_IN_FORMAT gives MATCH.format_code. Both are total on MATCH, so they are NOT NULL. PUBLISHES gives TEAM_RATING.published_by and PLAYER_RATING.published_by (FK to APP_USER, NOT NULL because ratings participate totally). The two binary PUBLISHES instances in the diagram are kept as two FKs.
- **M:N relationships (step 3):** HAS_STATUS_IN(team_id, format_code, status_since, status_end_date), REPRESENTS(player_id, team_id, from_date, to_date) and PARTICIPATES_IN(match_id, team_id, side,

total_runs, result). Relationship attributes sit on these relations. from_date is added to the PK of REPRESENTS because a player can return to a team, and the diagram records history by dates. For the same reason, status_since is added to the PK of HAS_STATUS_IN: a team can lose a format status and regain it later (for example, Zimbabwe left Test cricket in 2005 and returned in 2011), and status_end_date keeps the old period instead of deleting it. The exact-two-team and same-gender rules for PARTICIPATES_IN require separate integrity enforcement beyond the M:N foreign keys.

- **Multi-valued attributes (step 4): all separate relation:** playing_roles becomes PLAYER_ROLE(player_id, playing_role) with PK (player_id, playing_role); no fixed maximum is given in the requirements. retired_formats becomes PLAYER_RETIRED_FORMAT(player_id, format_code, retired_on) with PK (player_id, format_code); it has its own sub-attribute retired_on, so option (a) does not apply.
- **Ternary relationship (step 5):** APPEARED_FOR has FKs to PLAYER, MATCH and TEAM and carries all seven per-match figures. Since the diagram states that each (player, match) has exactly one team, the PK is (player_id, match_id) and team_id is a non-key, NOT NULL FK. The constraint that this team participates in the match can be enforced in the constraint phase with the composite FK (match_id, team_id) to PARTICIPATES_IN.
- **Specialization (step 6):** Option 8A (one relation for the superclass and one per subclass). MATCH holds common attributes; TEST_MATCH(match_id, scheduled_days, days_played) and LIMITED_OVERS_MATCH(match_id, overs_per_side, dls_applied) use match_id as both PK and FK. This fits a total, disjoint specialization defined by format.
- **Recursive relationship (step 7):** SUPERVISES becomes APP_USER.supervisor_id, a nullable FK to APP_USER.
- **Not applicable:** the diagram has no union types or other constructs.
- **Alternate candidate keys to declare as UNIQUE:** ICC_MEMBER (short_code), (name); TEAM (team_code), (member_id, gender); PLAYER (first_name, last_name, date_of_birth, gender); FORMAT (format_name); SERIES (series_name, start_date); MATCH (series_id, match_no); APP_USER (username), (email).

Stage A snapshot: after ER-to-relational mapping

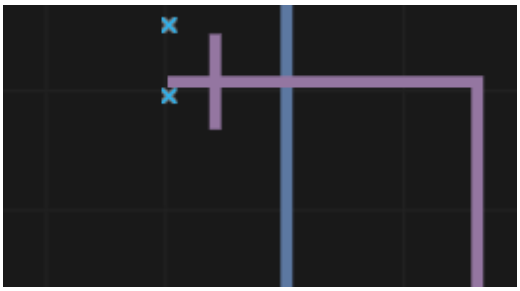


Link: https://drive.google.com/file/d/1kmLNVBB8iGGWLf3naACxm999m3iXAEdA/view?usp=sharing

Notation: Attributes listed vertically in each box. Underlined = primary key; italic = foreign key; arrows run from the foreign key to the primary key it references. Composite FKs are drawn with one arrow per column. Arrows are represented as crow's foot ends with



(ON FK SIDE)



(ON PK SIDE)

Stage B: First Normal Form (1NF)

No conversion needed. Every column in the Stage A model is atomic. The composite attributes full_name and venue were already split into single columns, and the multi-valued playing_roles and retired_formats are in their own relations (PLAYER_ROLE, PLAYER_RETIRED_FORMAT). No relation, including APPEARED_FOR and PARTICIPATES_IN, has a repeating group or a list in one column. Stage B therefore equals Stage A and no snapshot is submitted.

Stage C: Second Normal Form (2NF)

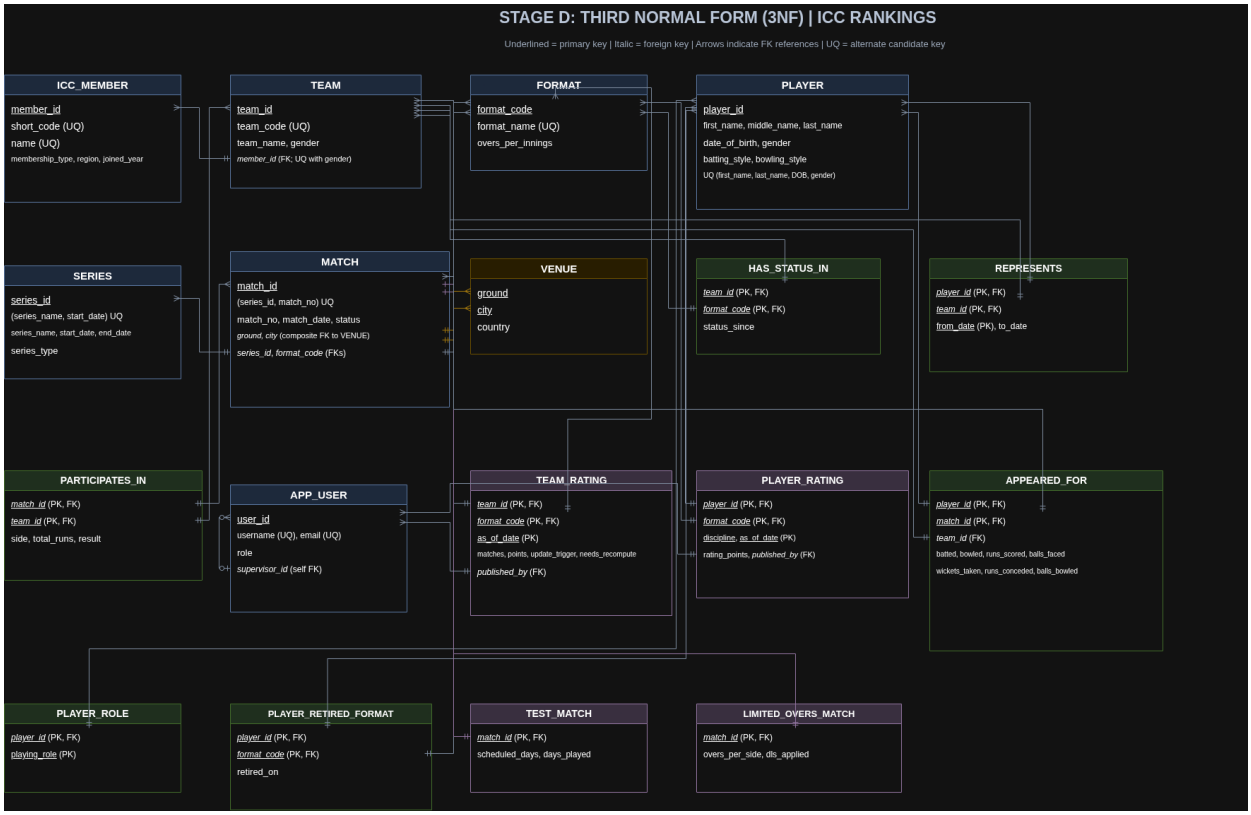
No conversion needed. Partial dependencies can only occur in relations with a composite candidate key, so those were checked against all their candidate keys. In HAS_STATUS_IN, REPRESENTS, PARTICIPATES_IN, APPEARED_FOR, TEAM_RATING, PLAYER_RATING and PLAYER_RETIRED_FORMAT, each non-key attribute (for example status_end_date, to_date, result, runs_scored, points, rating_points, retired_on) depends on the full key, not on part of it. PLAYER_ROLE is all-key. For the alternate composite keys (member_id, gender) in TEAM, (series_name, start_date) in SERIES, and (series_id, match_no) in MATCH, no non-prime attribute depends on only one of the columns: a member has both men's and women's teams, a series name recurs across years, and a series can contain matches of several formats. In PLAYER, the composite alternate key (first_name, last_name, date_of_birth, gender) was also checked: no subset of it determines middle_name, batting_style or bowling_style, because two different players can share a name or a birth date. All other primary keys are single-column. Stage C therefore equals Stage B and no snapshot is submitted.

Stage D: Third Normal Form (3NF)

One change: relation MATCH was split to create VENUE. In MATCH, the dependency ground, city → country holds by the meaning of the data: a named ground in a given city is in exactly one country. (ground, city) is not a superkey of MATCH, and country is non-prime, so country depends transitively on match_id. Storing country on every match repeats it for each match played at the same ground, which causes update anomalies. The fix is VENUE(ground, city, country) with PK (ground, city), and MATCH keeps ground and city as a composite FK to VENUE while country is removed. The determinant is (ground, city), not city alone, because city names repeat across ICC members: Hamilton has a ground in New Zealand (Seddon Park) and one in Bermuda (National Stadium). So city → country does not hold, and VENUE has no partial dependency.

The other relations were checked and are in 3NF. In PLAYER, the only determinants are player_id and the alternate key (first_name, last_name, date_of_birth, gender), both candidate keys, so no transitive dependency exists. For the listed relations, each determinant is a candidate key or superkey. TEST_MATCH and LIMITED_OVERS_MATCH have only match_id as determinant. APPEARED_FOR is in 3NF because team_id depends directly on (player_id, match_id), and the match figures do not determine one another.

Stage D snapshot: after conversion to 3NF (final relational model)

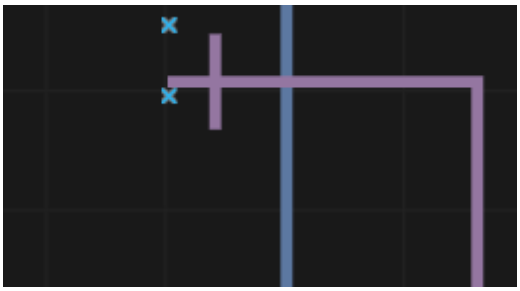


Link: https://drive.google.com/file/d/1NnWk8EaHMMfbQg6X7focqENXYthDmgRZ/view?usp=sharing

Notation: Attributes listed vertically in each box. Underlined = primary key; italic = foreign key; arrows run from the foreign key to the primary key it references. Composite FKs are drawn with one arrow per column. Arrows are represented as crow's foot ends with



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(ON PK SIDE)

Functional Dependencies and Candidate Keys

Dependencies come from the meaning of the ICC Rankings domain. Every relation was checked for Stages C and D; MATCH is the only one with a violation, and VENUE is the relation created from it.

Relation	**Candidate keys**	**Functional dependencies used**	**Result**
ICC_MEMBER	member_id; short_code; name	member_id → name, short_code, membership_type, region, joined_year; short_code → member_id; name → member_id	All determinants are keys. 3NF
TEAM	team_id; team_code; (member_id, gender)	team_id → team_code, team_name, gender, member_id; team_code → team_id; member_id, gender → team_id	All determinants are keys. 3NF
FORMAT	format_code; format_name	format_code → format_name, overs_per_innings; format_name → format_code	3NF
HAS_STATUS_IN	(team_id, format_code, status_since)	team_id, format_code, status_since → status_end_date	Full dependency. 3NF
PLAYER	player_id; (first_name, last_name, date_of_birth, gender)	player_id → first_name, middle_name, last_name, date_of_birth, gender, batting_style, bowling_style; first_name, last_name, date_of_birth, gender → player_id, middle_name, batting_style, bowling_style	Both keys checked. 3NF
PLAYER_ROLE	(player_id, playing_role)	No nontrivial functional dependency: every attribute is part of the candidate key.	All-key relation. 3NF
PLAYER_RETIRED_ FORMAT	(player_id, format_code)	player_id, format_code → retired_on	Full dependency. 3NF
REPRESENTS	(player_id, team_id, from_date)	player_id, team_id, from_date → to_date	3NF
SERIES	series_id; (series_name, start_date)	series_id → series_name, start_date, end_date, series_type; series_name, start_date → series_id	3NF
MATCH (before)	match_id; (series_id, match_no)	match_id → match_no, match_date, status, ground, city, country, series_id, format_code; series_id, match_no → match_id; ground, city → country	Violation: (ground, city) is not a superkey and country is non-prime.
MATCH (after)	match_id; (series_id, match_no)	match_id → match_no, match_date, status, ground, city, series_id, format_code; series_id, match_no → match_id	3NF
VENUE	(ground, city)	ground, city → country	3NF: the determinant is the candidate key.

Relation	Candidate keys	Functional dependencies used	Result
TEST_MATCH	match_id	match_id → scheduled_days, days_played	3NF
LIMITED_OVERS_MATCH	match_id	match_id → overs_per_side, dls_applied	3NF
PARTICIPATES_IN	(match_id, team_id)	match_id, team_id → side, total_runs, result	3NF
APPEARED_FOR	(player_id, match_id)	player_id, match_id → team_id, batted, bowled, runs_scored, balls_faced, wickets_taken, runs_conceded, balls_bowled	3NF
APP_USER	user_id; username; email	user_id → username, email, role, supervisor_id; username → user_id; email → user_id	3NF
TEAM_RATING	(team_id, format_code, as_of_date)	team_id, format_code, as_of_date → matches, points, update_trigger, needs_recompute, published_by	3NF
PLAYER_RATING	(player_id, format_code, discipline, as_of_date)	player_id, format_code, discipline, as_of_date → rating_points, published_by	3NF

Changes Made to the Assigned EER Diagram

No entity, attribute or relationship of the assigned diagram was added or removed. The following mapping decisions go slightly beyond a literal reading and are listed for transparency.

- REPRESENTS gets from_date in its primary key.

  Reason: the diagram stores from_date and to_date to keep history, and (player_id, team_id) alone would block a player from rejoining a team.

- APPEARED_FOR uses PK (player_id, match_id) rather than all three FKs.

  Reason: the diagram states that each (player, match) has exactly one team. The rule that team_id must be one of the two teams of that match (composite FK to PARTICIPATES_IN) is left for the constraint phase.

- The requirement conflict noted in Team 41's report (total participation of MATCH in APPEARED_FOR versus appearances allowed only for Completed matches) is not resolved in the relational model; no constraint is placed on MATCH, so matches without appearances are allowed.

- HAS_STATUS_IN gets status_since in its primary key, giving (team_id, format_code, status_since).
  Reason: the diagram stores status_since and status_end_date to keep history, and (team_id, format_code) alone would block a team from regaining a status it lost earlier.

- format_code in TEAM_RATING, PLAYER_RATING and PLAYER_RETIRED_FORMAT is declared as a foreign key to FORMAT, although the EER shows it only as an attribute with no relationship to FORMAT.

  Reason: it holds the same codes (TEST, ODI, T20I), and the FK stops an unknown format from being stored.

- Stage D introduces VENUE. This is a normalization result, not an EER change.

GenAI Usage Statement

We used Claude to produce a plan of action and step by step procedure using Team 41's Phase 2 report and Team 45's Phase 1 requirements. We also used ChatGPT to review this draft against the Phase 3 brief and the supplied source documents, identify inconsistencies and missing checks, and make editorial and structural corrections.